

Erdős Problem #86

EP-86

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let Q_n be the n -dimensional hypercube graph (so that Q_n has 2^n vertices and $n2^{n-1}$ edges). Is it true that every subgraph of Q_n with $\geq \left(\frac{1}{2} + o(1)\right)n2^{n-1}$ many edges contains a C_4 ?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>